



LINEAR ALGEBRA

UE19MA251

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Agenda



- Least square problem – Normal equations.
- Positive Definiteness and the Least squares
- A simple example

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The Normal Equations



Theorem: *Any solution x of the least squares problem is a solution of the linear system*

$$A^T A x = A^T b.$$

The system is nonsingular if and only if A has linearly independent columns.

Proof. ● Since $b - Ax \in \ker(A^T)$, we have $A^T(b - Ax) = 0$ or
 $A^T A x = A^T b$.

● $A^T A$ is nonsingular. Suppose $A^T A x = 0$ for some $x \in \mathbb{R}^n$. Then
 $0 = x^T A^T A x = (Ax)^T Ax = \|Ax\|_2^2$. Hence $Ax = 0$ which implies that
 $x = 0$ if and only if A has linearly independent columns.

The linear system $A^T A x = A^T b$ is called the **normal equations**.

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Positive Definite Matrices and Least Squares



We have learned that **least square** comes from **projection** :

$$b - p = e \Rightarrow, A^T(b - A\hat{x}) = 0 \Rightarrow, A^T A\hat{x} = A^T b$$

Consequently, only if $A^T A$ is invertible, then we can use linear regression to find approximate solutions $\hat{x} = (A^T A)^{-1} A^T b$ to unsolvable systems of linear equations.

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According to the reasoning before, we know as long as all columns of $A_{m \times n}$ are mutual independent, then $A^T A$ is invertible. At the same time we ought to notice that the columns of A are guaranteed to be independent if they are orthogonal and even orthonormal.

In another prospective, if $A^T A$ is positive definite, then $A_{m \times n}$ has rank n (independent columns) and thus $A^T A$ is invertible.

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Positive Definite Matrices and Least Squares



Overall, if $A^T A$ is positive definite or invertible, then we can find approximate solutions of least square.

Find the function of the form

$$F(x) = c_1 + c_2 x \lg x + c_3 e^x$$

that is the best least-squares fit to the data points

$$(1, 1), (2, 1), (3, 3), (4, 8).$$

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Positive Definite Matrices and Least Squares



First we form the A matrix

$$A = \begin{pmatrix} 1 & 0 & e \\ 1 & 2 & e^2 \\ 1 & 3 \lg 3 & e^3 \\ 1 & 8 & e^4 \end{pmatrix}$$

We compute the pseudoinverse, then multiply it by y , to obtain the coefficient vector

$$c = \begin{pmatrix} 0.411741 \\ -0.20487 \\ 0.16546 \end{pmatrix}.$$

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Positive Definite Matrices and Least Squares



- Given $A^{m,n}$ and $b \in \mathbb{R}^m$.
- The system $Ax = b$ is **over-determined** if $m > n$.
- This system has a solution if $b \in \text{span}(A)$, the column space of A , but normally this is not the case and we can only find an approximate solution.
- A general approach is to choose a vector norm $\|\cdot\|$ and find x which minimizes $\|Ax - b\|$.
- We will only consider the Euclidian norm here.

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Positive Definite Matrices and Least Squares



- Given $A^{m,n}$ and $b \in \mathbb{R}^m$ with $m \geq n \geq 1$. The problem to find $x \in \mathbb{R}^n$ that minimizes $\|Ax - b\|_2$ is called the **least squares problem**.
- A minimizing vector x is called a **least squares solution** of $Ax = b$.
- Several ways to analyze:
 - Quadratic minimization
 - Orthogonal Projections
 - SVD

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- Define function $E : \mathbb{R}^n \rightarrow \mathbb{R}$ by $E(x) = \|Ax - b\|_2^2$
- $E(x) = (Ax - b)^T (Ax - b) = x^T Bx - 2c^T x + \alpha$, where
- $B := A^T A$, $c := A^T b$ and $\alpha := b^T b$.
- B is positive semidefinite and positive definite if A has rank n .
- Since the Hessian $HE(x) := \left(\frac{\partial^2 E(x)}{\partial x_i \partial x_j} \right) = 2B$ we can find minimum by setting partial derivatives equal zero.
- $\nabla E(x) := \left(\frac{\partial E(x)}{\partial x_i} \right) = 2(Bx - c) = 0$
- **Normal equations** $A^T Ax = A^T b$.

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A simple example

$$\begin{array}{l} x_1 = 1 \\ x_1 = 1, \\ x_1 = 2 \end{array} \quad \mathbf{A} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \quad \mathbf{x} = [x_1], \quad \mathbf{b} = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix},$$

- Quadratic minimization problem:

$$\|\mathbf{Ax} - \mathbf{b}\|_2^2 = (x_1 - 1)^2 + (x_1 - 1)^2 + (x_1 - 2)^2.$$

- Setting the first derivative with respect to x_1 equal to zero we obtain
 $2(x_1 - 1) + 2(x_1 - 1) + 2(x_1 - 2) = 0$ or $6x_1 - 8 = 0$ or $x_1 = 4/3$
- The second derivative is positive (it is equal to 6) and $x = 4/3$ is a global minimum.



THANK YOU

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